



Prediction of Online Shoppers Intention using Ensemble Learning Model after Treating Imbalance and Outlier Data

B.Shathya

Department of BCA,
Ethiraj college for women,
Chennai, India

shathya04@gmail.com

Published online: September, 2025

Abstract— Data mining is also known as knowledge discovery is the process of uncovering patterns and extracting knowledge from large dataset. Data mining offers many applications in business. It helps a company to increase sales from the behaviour and practices of its customers. It pulls out information from data sets and compares it to help the business to make decisions. Classification is one of the data mining techniques used to assign a class label to each instance in a dataset. There are many classification techniques available. The performance of each model will vary based on the dataset and data mining techniques used. The proposed work is evaluated using Online Shoppers Purchasing Intention Dataset from UCI Machine Learning Repository website. This binary classification dataset is imbalanced with an unequal class distribution and many data points that fall on outside the pattern. The imbalance dataset and outliers will create bias in the result. To overcome this problem the outliers in the dataset are removed using Z-Score method. The important features are selected using Recursive Feature Elimination method and the imbalanced dataset is balanced using the oversampling technique SVM-SMOTE. The online shopper's intention is predicted using Machine Learning algorithms. The individual models like K-Nearest Neighbor, Support Vector Machines, Random Forest, Naïve Bayes have been built. The popular ensemble methods like boosting, bagging and stacking have been built. The proposed stacking model has been built using K-Nearest Neighbor, Support Vector Machines, Random Forest, Naïve Bayes as base model. By using the output of the base models, the meta model is trained. The Performance measures of all the models are found. The analysis results show that Proposed stacking ensemble classification leads improved performance.

Keywords—Classification, Ensemble Learning, Recursive Feature Elimination, SVM-SMOTE, Outlier, Z-Score

I. INTRODUCTION

Online shopping is a part of E-Commerce which allows the customer buys through a digital platform. It allows customers to directly buy goods and services through internet without going to the physical shop. The number of online shopping consumers and online shopping revenue has continuously increased over the years [1]. Online shopping intention refers to the probability that a consumer make online shopping in future. It is important to predict the purchasing intention of online shoppers by classification techniques. To find out the best technique requires comparison between them. Before applying the correct classification technique in dataset, the quality of the data should be checked. The online shopper's dataset is an imbalanced dataset with many outliers. If any data points are higher or lower than others it may be treated as outlier. The Box plot is a method for graphically displaying the distribution of data to easily detect outliers. After analysing the outliers in each attribute, the outliers are eliminated using Z-Score method. Feature selection refers to a technique that selects the most relevant features for a dataset. Fewer features can allow the classification algorithms to run more efficiently and effectively. The irrelevant input features produce worst performance. After removing outliers, the Recursive Feature Elimination feature selection algorithm is used to eliminate weakest features from a training dataset. Then the imbalance dataset transformed to balanced dataset using SVM-SMOTE which is one of the Synthetic Minority Over-sampling techniques (SMOTE) techniques. The various classification techniques like K-Nearest Neighbor, Support Vector Machines, Random Forest, Naïve Bayes and the three types of ensemble learning methods have been conducted to find out the most suitable technique. The performance of classification techniques is compared by Performance evaluation measures.

II. RELATED WORK

Chien Chin Chen et al. [2] investigated the factors that affect buyers' decisions and predicted the success of online group buyers' decisions. They utilized five dimensions and thirteen features derived from previous research to demonstrate the efficacy of the proposed method. Irina Rish [3] demonstrated that naive Bayes works well for certain functional feature dependencies and found that the accuracy of naive Bayes is not directly correlated with the degree of feature dependencies. Abdul Rasheed [4] proposed wide variety of techniques and methods to assess the importance of feature selection out of the available features. The results were discussed from the obtained features and the selected features with respect to the method chosen. Bissan Ghaddar [5] addressed the problem of feature selection within support vector machine classification that deals with finding an accurate binary classifier that uses a minimal number of features which are the key for the construction of advanced decision support system. Neha Sharma [6] discussed the classification process using the Naive Bayes Classifier and the recent developments in the same particularly in domains of health care and semantic analysis. Durgesh et al. [7] obtained support vectors for classification by learning from the training samples and showed the comparative results using different kernel functions for all data samples.

Marta Baldomero Naranjo et al. [8] proposed a robust classification model, based on support vector machine (SVM), which simultaneously deals with outlier detection and feature selection. The classifiers obtained with the heuristic method are tested and compared with existing SVM-based models to demonstrate their efficiency. Sondas Jameel Mukhyber et al. [9] investigated the impact of Z-Score normalization method on accuracy of classification of liver disease and observed the result of classification algorithms effected with Z-Score normalization. Boen Yang et al. [10] demonstrated an ensemble learning based housing price prediction model to improve the predicting accuracy and stability incorporating various predictors like random forest, GBDT and XGB algorithms. Mohd Khalid Awang et al. [11] proposed a stacking ensemble based on six different learning algorithms as the base classifiers and tested on five different meta model classifiers. The performance of the proposed stacking ensemble model compared with single classifiers, bagging and boosting ensemble.

III. MATERIALS AND METHODS

i. Outlier Detection

The values that significantly separate from all other values are termed as outliers. Outliers in the dataset may cause problems during model fitting. A box plot or whisker plot is a chart used to visually show the distribution of numerical data. It identifies mean values and the dispersion of the data. The extreme values are denoted by bubbles are outliers in the dataset. Figure 1 shows the boxplot diagram for ProductRelated attribute. Figure 2 shows the boxplot diagram for ExitRates, BounceRates attributes. Figure 3 shows the boxplot diagram for ProductRelated_Duration attribute. By using boxplot diagram method ProductRelated, ExitRates, BounceRates, ProductRelated_Duration attributes are found with more than 300 outliers.

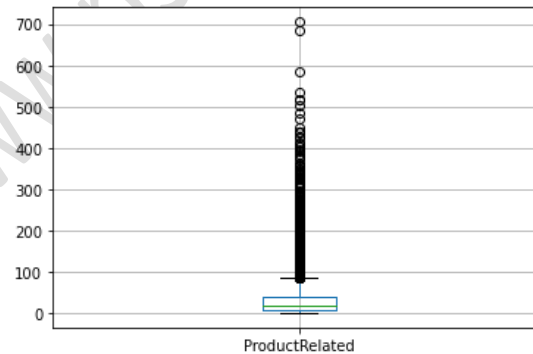


Fig 1. Boxplot for ProductRelated attribute

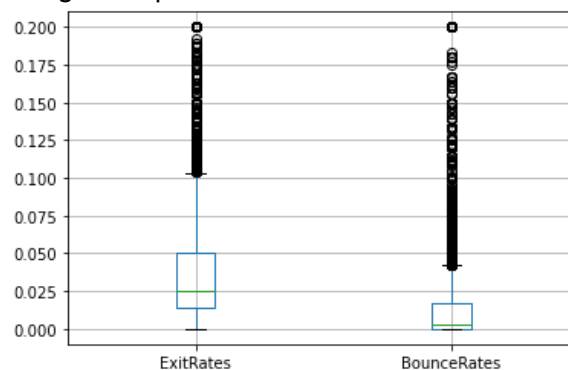


Fig 2. Boxplot for ExitRates, BounceRates attributes

Z-Score is one of the popular techniques to determine outliers. If the Z-Score value of a data point is less than -3 or greater than +3, the data point will be treated as outlier.

$$\text{Z-Score} = (\text{Observation} - \text{Mean}) / \text{Standard Deviation} \quad (1)$$

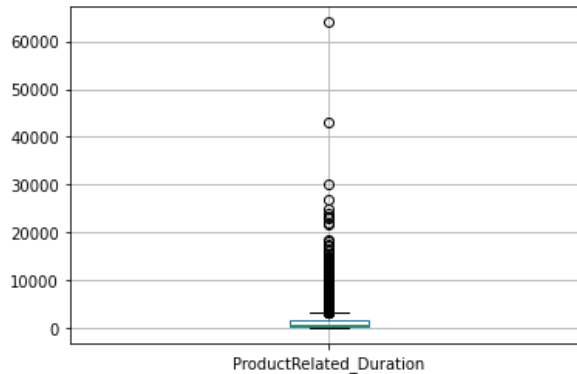


Fig 3. Boxplot for ProductRelated_Duartion attribute

Equation (1) is used to calculate Z-Score. The dataset after removing outliers are taken for further study.

ii. Feature Selection

Feature selection or attribute selection is the process of reducing the features to be used for training a model. Some features might be unrelated to the task and have no relation with the target variable. Discarding irrelevant features will prevent the model from overfitting.

```
Index(['Administrative', 'Administrative_Duration', 'ProductRelated',
      'ProductRelated_Duration', 'BounceRates', 'ExitRates', 'PageValues',
      'Month', 'Revenue'],
      dtype='object')
```

Fig 4: Attributes after Feature Selection

Recursive feature elimination (RFE) method is a wrapper type forward feature selection method that is used to select features by recursive method from training dataset. It removes the weakest feature until the desired number of features are achieved. Figure 4 shows the selected optimal number of attributers after removing unnecessary features.

iii. Handling imbalanced dataset

Imbalanced dataset refers to the dataset with the unequal distribution of classes in the training dataset. SVM-SMOTE is an oversampling method to treat imbalance dataset. SVM-SMOTE generates new minority class instances near borderlines with Support Vector Machine to establish boundary between classes. SVM-SMOTE is used to generate balanced dataset.

iv. Classification methods

1. K-Nearest Neighbor: The K-Nearest Neighbor(KNN) is an effective non-parametric supervised learning classifier. It does not make any presumptions on the elementary data. A labelled training dataset is provided where the data points are categorized into various classes to predict the class of the unlabelled data [12].

2. Support Vector Machines: Support vector machines are powerful supervised machine learning algorithms generally used in classification problems. It handles multiple continuous and categorical variables. The classification happens by determining a hyperplane after all the data points are plotted.

3. Random Forest: A random forest is a supervised machine learning algorithm that is constructed from decision tree algorithms to solve classification as well as regression problems. The greater number of trees overcome the problem of overfitting. It takes less training time compared to other models.

4. Naive Bayes: The Naive Bayes classifiers are based on the Bayes Theorem and are referred to as Bayesian classifiers. Naive Bayes classifiers work on an independent assumption that the input features participating in the classification are totally independent. The classification models have been created by using the above four classifiers.

v. Ensemble Methods

A key concept of the ensemble technique is that it seeks to combine ideas from many individual classifiers in order to get superior results that complement one another [13].

Bagging is also known as Bootstrap aggregation. It is an ensemble learning method improves accuracy by reducing variance within noisy dataset. It takes random subsets and fits a classifier to each subset. The accuracy of bagging classifier is found.

Boosting is an ensemble technique used to reduce bias in predictive data analysis. It combines multiple weak learners sequentially to make a strong learner.

The performance of all the boosting algorithms like Adaboost, Gradient Boosting algorithm, Extreme Gradient Boosting algorithm, LightGBM, CatBoost also found to compare.

Stacking or Stacked Generalization is an ensemble machine learning algorithm that uses a meta-learning algorithm to combine the predictions from two or more base machine learning algorithms. stacking often takes into account heterogeneous weak learners, while bagging and boosting mostly take into account, homogeneous weak learners. Second, stacking uses a meta-model to combine the basic models, while bagging and boosting use stochastic methods to combine weak learners.

In this research paper, the four heterogeneous classifiers are used to create Level 1 Base classifiers. The selected individual base models are as follows:

Model 1 : K-Nearest Neighbor

Model 2 : Support Vector Machines

Model 3 : Random Forest

Model 4 : Naïve Bayes

The output of the Level 1 base classifiers is then used to train a Level 2 meta classifier.

IV. RESULT ANALYSIS

The Online Shoppers Purchasing Intention Dataset is acquired from UCI Machine Learning Repository website [14]. The dataset contains 18 attributes and 12330 sessions. Each session belongs to a different user in a one-year period. In these 12330 sessions, 1908 sessions are positive samples ending with shopping and 10422 sessions are negative samples that did not end with shopping. There are 10 numerical and 8 categorical attributes in the dataset. The dataset with an unequal class distribution is imbalanced. In this imbalanced binary classification problem, the Revenue attribute is used as the class label.

By using Z-Score method totally 2774 outliers are detected and removed. After removing outliers, there are 9556 sessions in the dataset. RFE removes the 10 weakest features to improve the prediction. The remaining 8 features are used in creating model. After balancing dataset using SMOTE variant, the dataset contains 16372 samples. The balanced data set contains 8186 positive and 8186 negative data. Now the data set has been divided into 80% for training (13097 sessions) and 20% for testing (3275 sessions) purpose.

To examine the performance of the classification algorithms, the performance measure accuracy, Precision, Recall, F1 Score and Jaccard Score are used. Table 1 shows the various performance metrics of base classification models.

Model	Accuracy	Precision	Recall	F1 Score	Jaccard Score
K-Nearest Neighbors	0.887	0.862	0.922	0.891	0.80
Support Vector Machine	0.915	0.910	0.921	0.916	0.85
Random Forest	0.923	0.916	0.944	0.924	0.87
Naïve Bayes	0.857	0.888	0.818	0.852	0.74

Table 1 : Performance Measures of Base classifiers

Accuracy is a metric that generally describes how the model performs across all classes and also tells the percentage of correct predictions made by a model. It is useful when all classes are of equal importance.

Accuracy = Number of Correct Predictions/Total Number of Predictions

(2)

Jaccard Index or Jaccard similarity coefficient is used to calculate and find out the accuracy of a classification model. The Higher value of the Jaccard index shows the higher accuracy of the classifier.

Ensemble Technique	Accuracy	Precision	Recall	F1 Score	Jaccard Score
AdaBoost	0.915	0.910	0.923	0.916	0.85
Gradient Boosting	0.920	0.910	0.937	0.921	0.85
Extreme Gradient Boosting	0.920	0.905	0.939	0.922	0.85
LightGBM	0.930	0.918	0.947	0.932	0.87
Bagging	0.931	0.921	0.945	0.933	0.87
CatBoost	0.933	0.922	0.950	0.934	0.88
Stacking	0.940	0.928	0.950	0.939	0.89

Table 2 : Performance Measures of Ensemble classifiers

The accuracy of the individual model Random Forest classifier outperformed well with 92.3% of accuracy and 92.4% of F1 Score compare with other models. Table 2 shows the various performance metrics of ensemble

learning techniques. All the boosting algorithms performs better than K-Nearest Neighbor. The various boosting classifiers produce the accuracy and F1 score better than K-Nearest Neighbor, Support Vector Machines, Naive Bayes. In the boosting algorithms, the CatBoost algorithm outperform well with the high accuracy of 93.3% and the AdaBoost algorithm produces the low accuracy of 91.5%. But the Bagging classifier produces the model with the accuracy of 93.1% and Jaccard score 0.87 which is better than all the boosting classifiers. The Proposed stacking algorithm with the base models (K-Nearest Neighbor, Support Vector Machines, Random Forest, Naive Bayes) has a high performance with the accuracy of 94% and F1 score of 93.9% since the base classifiers are stacked.

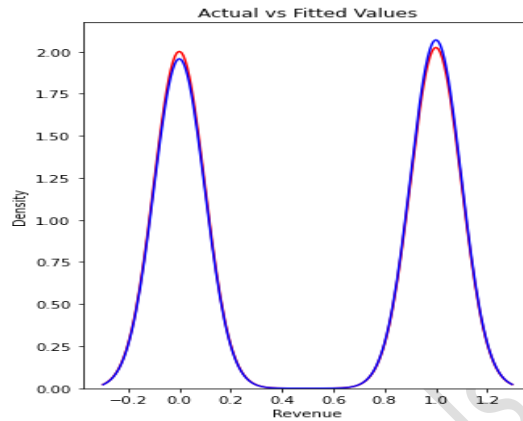


Fig 4. Visualization of accuracy of Proposed Stacking ensemble model

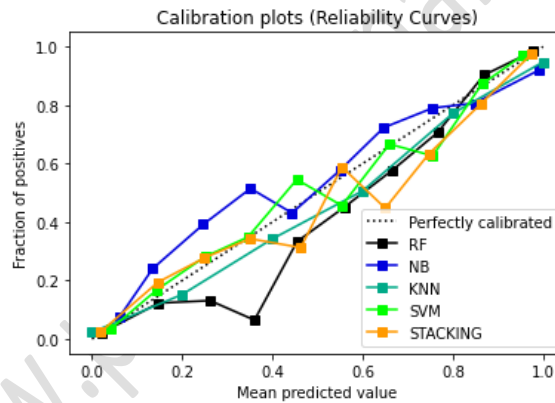


Fig 5. Calibration Plot for Base and Proposed Stacking ensemble model

The Proposed stacking algorithm has a highest Jaccard score of 0.89. Figure 4 shows the distplot curve that visualize the distribution of actual and fitted accuracy of proposed stacking ensemble model. Figure 5 shows the calibration curve for all base models and proposed stacking ensemble model to check if the predicted value can directly be interpreted as confidence level. It works well for binary classification.

V. CONCLUSION

In this study, the four different classification algorithms K-Nearest Neighbor, Support Vector Machines, Random Forest, Naive Bayes, Bagging and Boosting algorithms Adaboost, Gradient Boosting algorithm, Extreme Gradient Boosting algorithm, LightGBM, CatBoost are investigated. As many research work are conducted in ensemble models, different classifiers are used as base level. In this study, the proposed ensemble stacking model with the base models K-Nearest Neighbor, Support Vector Machines, Random Forest, Naive Bayes outperformed well when compared with single classifiers, bagging and boosting ensemble for Predicting Online Shoppers Intention.

References

- [1] H.Ozen and N.Engizek, "Shopping online without thinking: being emotional or rational?", Asia Pacific Journal of Marketing and Logistics, 2014, pp.78-93.
- [2] Chien Chin Chen, Meng-Chieh Chung, "Predicting the success of group buying auctions via classification", Knowledge-Based Systems, Volume 89, 2015, pp.627-640.
- [3] Rish, Irina, "An Empirical Study of the Naïve Bayes Classifier", IJCAI 2001, Work Empir Methods Artif Intell, pp41-46, Jan 2001.
- [4] A. Abdul Rasheed, "Feature Selection: An Assessment of Some Evolving Methodologies", Turkish Journal of Computer and Mathematics Education, Vol. 12, 2021.

- [5] Bissan Ghaddar, Joe Naoum-Sawaya, "High Dimensional Data Classification and Feature Selection using Support Vector Machines", European Journal of Operational Research, 2017.
- [6] Neha Sharma, "Classification Using Naive Bayes- A Survey", International Journal of Engineering Science Invention Research & Development, Vol. 2, February 2016.
- [7] Srivastava, Durgesh & Bhambhu, Lekha, "Data classification using support vector machine", Journal of Theoretical and Applied Information Technology, 2010, pp.1-7.
- [8] Marta Baldomero-Naranjo, Luisa I. Martínez-Merino, Antonio M. Rodríguez-Chía, "A robust SVM-based approach with feature selection and outliers detection for classification problems", Expert Systems with Applications, Volume 178, 2021.
- [9] Sondos Jameel Mukhyber, Dhahir Abdulhade Abdulah, Amer D. Majeed, "Effect Z-Score Normalization on Accuracy of classification of liver disease", Turkish Journal of Computer and Mathematics Education, Vol.12, 2021, pp.658- 662.
- [10] Bowen, Y. , Buyang, C. , "Research on Ensemble Learning-based Housing Price Prediction Model", Big Geospatial Data and Data Science, 2018.
- [11] Mohd Khalid Awang, Mokhairi Makhtar, Norlina Udin and Nur Farraliza Mansor, "Improving Customer Churn Classification with Ensemble Stacking Method", International Journal of Advanced Computer Science and Applications, 2021.
- [12] Taunk, Kashvi & De, Sanjukta & Verma, Srishti & Swetapadma, Aleena, "A Brief Review of Nearest Neighbor Algorithm for Learning and Classification", 2019, pp. 1255-1260.
- [13] M. Mohamad, M. Y. M. Saman, and N. A. Hamid, "Complexity Approximation of Classification Task for Large Dataset Ensemble Artificial Neural Networks", Lect. Notes Electr. Eng., vol. 520, April 2019, pp. 195-2019.
- [14] Sakar.C.O., Polat.S.O., Katircioglu.M, et al., "Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks", Neural Computing & Applications, 2018, pp.6893-6908.



This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)